

Major open source solutions powering MakeMyTrip

- Apache Hadoop, Storm and Spark for Big Data analytics
- Guava from Google and CoreNLP from Stanford for machine learning
- Jenkins, OpenTSDB, Grafana, and the ELK stack for production deployment and monitoring pipeline
- Apache HTTPD, Tomcat, NGINX, Django and MySQL for production infrastructure

“We were leveraging machine learning for Web anomaly detection and other use cases, when we found the requirement for a framework that would help security teams to quickly write their own use cases with minimal effort. That brought dataShark to life,” Mehta told Open Source For You.

Challenges in building a community solution

Building dataShark for the community involved some challenges for the team. “Being in the security field, it was challenging to up-skill ourselves to Big Data systems and machine learning,” says Kunal Aggarwal, senior information security engineer, MakeMyTrip.



MakeMyTrip team while taking a hands-on experience on open source technologies

Aggarwal is among the four-member team, which is led by Mehta and also includes Security Analyst Avinash Jain and Senior Security Analyst Dhruv Kalaan alongside Mehta and Aggarwal himself, tasked with maintaining the dataShark repository on GitHub, and with updating the open source project with new upgrades. Despite being a small team, the members preferred to utilise the inhouse skillsets to make dataShark successful, before releasing it in the public. “Once we had successfully acquired the skillsets, the journey with dataShark has been a breeze,” says Aggarwal.

Moving on from proprietary solutions

Mehta’s team is actively developing strategies to move from the commercialised proprietary world to the fast-growing open



Open source technologies help enhancing web traffic of MakeMyTrip

source ecosystem. The MakeMyTrip website, which receives over 22 million visits in a month, has recently shifted from some Microsoft technologies to open source implementations. “At an optimal operating expenditure, we were able to exercise greater flexibility using open source components on our website,” Mehta states.

MakeMyTrip is also considering open source as a parameter to find the right talent. The company believes that it is important for young professionals to opt for open source skills. “Open source platforms nowadays have become mainstream in the consumer Internet space, and the new-age software professional has to consider exposure to these platforms,” Mohan told Open Source For You.

Security enhancements through open source

dataShark is one of the publicly released examples of how MakeMyTrip is using open source to secure the user experience. However, there are several other community-backed options it uses on the security front. “We leverage various open source deployments to manage security. These have greatly helped to enhance our security posture in the market,” says Mehta.

Open source: An efficient operating model

Like many other open source adopters, the IT experts at MakeMyTrip are highly satisfied with the open source success rate. “The use of open source software not only reflects well on the technical expertise of the organisation, but is also a positive and efficient operating model,” says Mohan.

The CTO concludes that MakeMyTrip will continue to leverage open source on a large scale in the future and contribute to the community wherever possible. 

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